

Problem 17.1.11

Sketch or graph the given region and its image under the given mapping.

$$|z| \leq \frac{1}{2}, \quad -\frac{\pi}{8} < \operatorname{Arg} z < \frac{\pi}{8}, \quad w = z^2$$

Solution

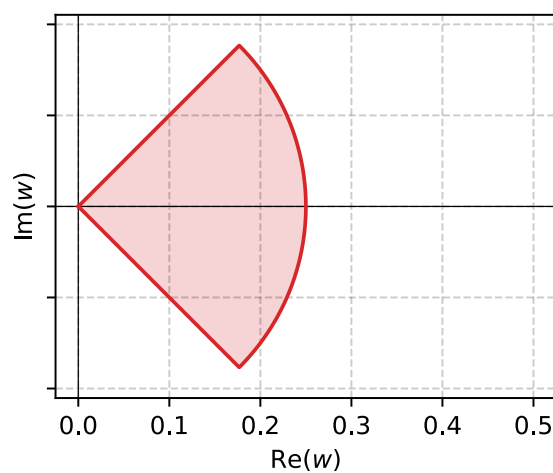
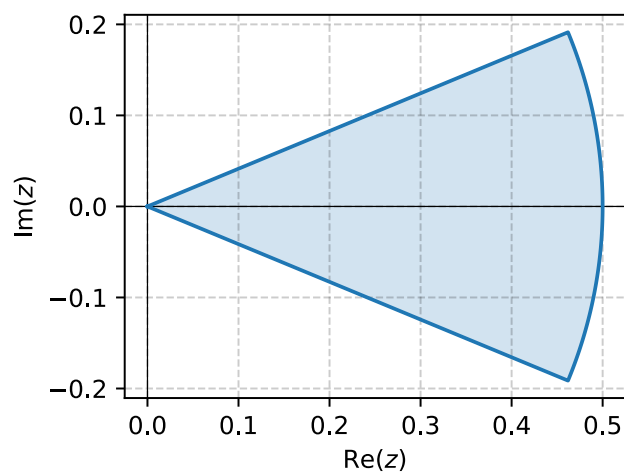
$$z = re^{i\theta}$$

$$w = Re^{i\phi}$$

$$z^2 = r^2 e^{2i\theta}$$

$$R = r^2 = \frac{1}{4}$$

$$\phi = 2\theta \implies -\frac{\pi}{4} < \operatorname{Arg} w < \frac{\pi}{4}$$



Problem 17.1.13

Sketch or graph the given region and its image under the given mapping.

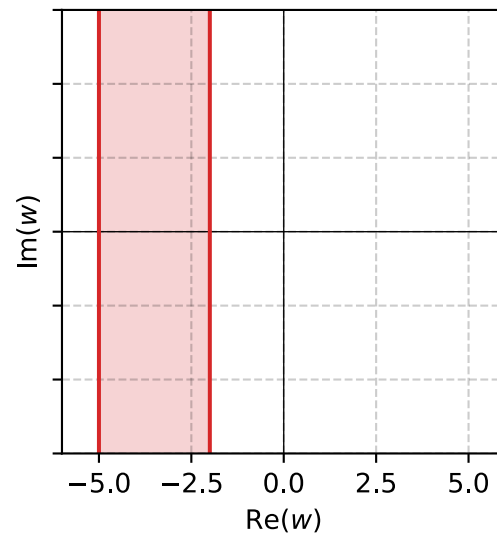
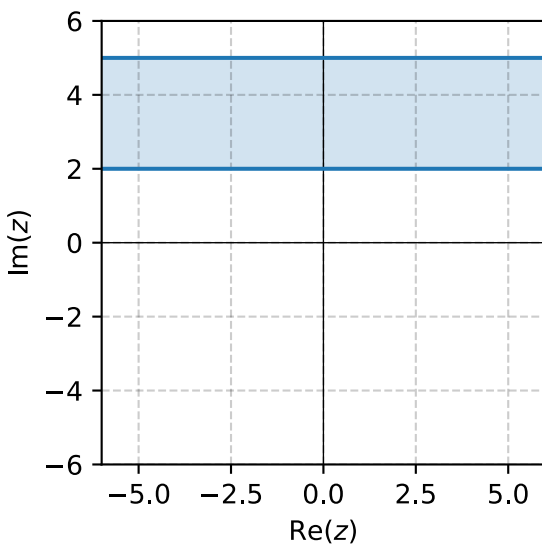
$$2 \leq \operatorname{Im} z \leq 5, \quad w = iz$$

Solution

The region is an infinite strip bounded by $z_1 = 2i$ and $z_2 = 5i$, mapping to $w = iz$ gives another infinite strip bounded by

$$w_1 = iz_1 = -2$$

$$w_2 = iz_2 = -5i$$



Problem 17.1.15

Sketch or graph the given region and its image under the given mapping.

$$\left| z - \frac{1}{2} \right| \leq \frac{1}{2}, \quad w = \frac{1}{z}$$

Solution

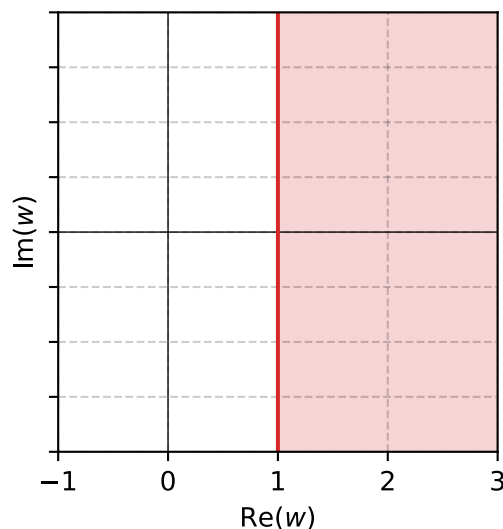
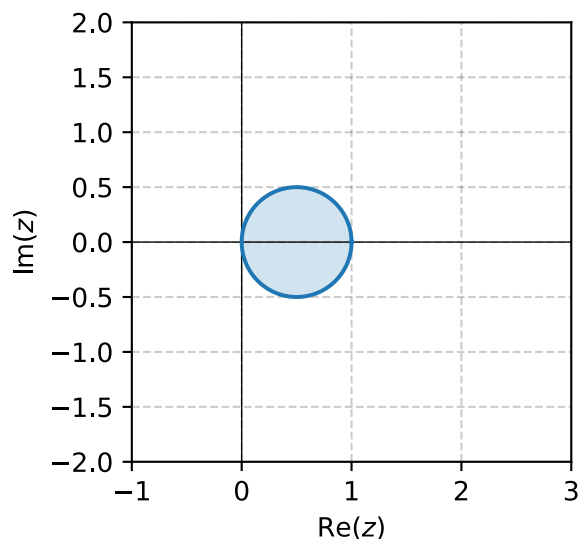
The boundary of the region $z = x + iy$ has the equation

$$\begin{aligned} \left(x - \frac{1}{2} \right)^2 + y^2 &= \left(\frac{1}{2} \right)^2 \\ x^2 - x + \frac{1}{4} + y^2 &= \frac{1}{4} \\ x^2 - x + y^2 &= 0 \\ x^2 + y^2 = x &\implies |z|^2 = \operatorname{Re} z \end{aligned}$$

Applying this relation to $w = u + iv = \frac{1}{z}$ we can find the boundary of the mapping.

$$\begin{aligned} \left| \frac{1}{w} \right|^2 &= \operatorname{Re} \frac{1}{w} \\ \frac{1}{|w|^2} &= \operatorname{Re} \frac{1}{u + iv} \\ \frac{1}{|w|^2} &= \operatorname{Re} \frac{u - iv}{u^2 + v^2} \\ \frac{1}{|w|^2} &= \frac{u}{|w|^2} \\ u &= 1 \end{aligned}$$

This is a vertical line. Since $z_c = \frac{1}{2}$ is within the original region, then $w_c = \frac{1}{z_c} = 2$ which is on the right side of the boundary and therefore the mapped region is the half plane on the positive side of $u = 1$



Problem 17.1.17

Sketch or graph the given region and its image under the given mapping.

$$\operatorname{Ln} 2 \leq x \leq \operatorname{Ln} 4, \quad w = e^z$$

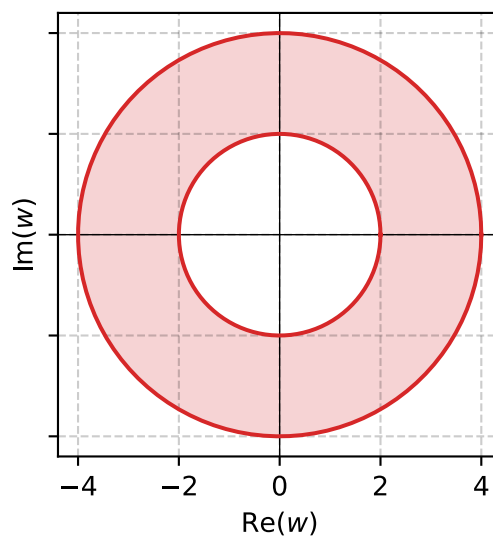
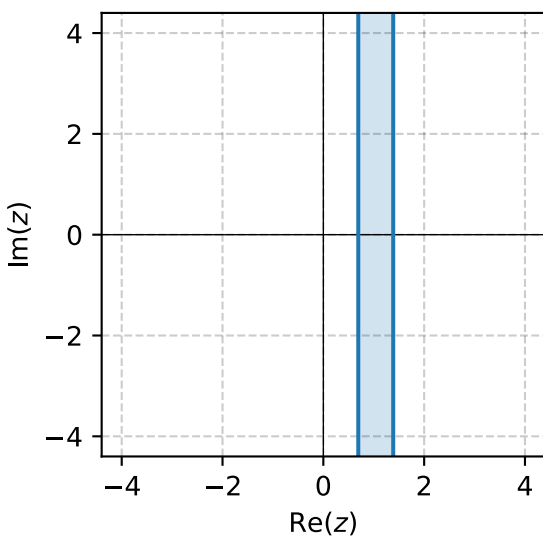
Solution

The region is an infinite vertical strip bounded by $z_1 = \operatorname{Ln} 2$ and $z_2 = \operatorname{Ln} 4$,

$$|w_1| = |e^{z_1}| = |e^{\operatorname{Ln} 2}| = 2$$

$$|w_2| = |e^{z_2}| = |e^{\operatorname{Ln} 4}| = 4$$

The new boundary is two concentric circles.



Problem 17.1.19

Sketch or graph the given region and its image under the given mapping.

$$1 < |z| < 4, \quad \frac{\pi}{4} < \theta \leq \frac{3\pi}{4}, \quad w = \operatorname{Ln} z$$

Solution

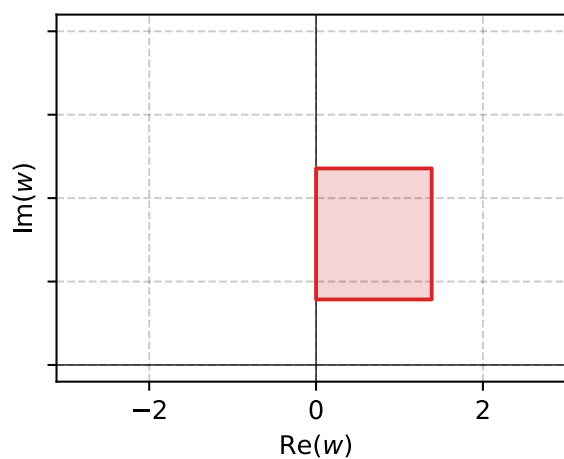
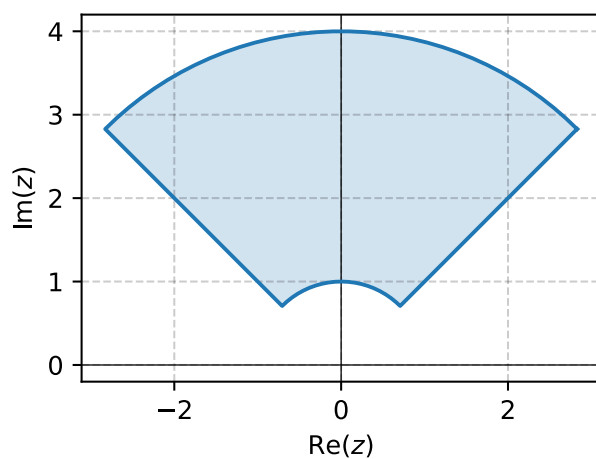
The region is bounded by 4 borders that can be described in the form $z = re^{i\theta}$.

Mapping to

$$\begin{aligned} w &= \operatorname{Ln} z \\ &= \ln |z| + i \operatorname{Arg} z \\ &= \ln r + i\theta \\ &= u + iv \end{aligned}$$

$$u = r \implies \ln 1 < u < \ln 4$$

$$v = \theta \implies \frac{\pi}{4} < v \leq \frac{3\pi}{4}$$



Problem 17.2.7

Find the inverse $z = z(w)$ Check by solving $z(w)$ for w .

$$w = \frac{i}{2z - 1}$$

Solution

$$\begin{aligned} w &= \frac{i}{2z - 1} \\ 2z - 1 &= \frac{i}{w} \\ z &= \frac{i}{2w} + \frac{1}{2} \end{aligned}$$

$$\boxed{z(w) = \frac{i}{2w} + \frac{1}{2}}$$

Problem 17.2.13

Find the fixed points.

$$w = 16z^5$$

Solution

$$f(z) = z$$

$$w = 16z^5 = z$$

$$16z^5 - z = 0$$

$$z(16z^4 - 1) = 0 \implies \boxed{z = 0, \pm \frac{1}{2}, \pm \frac{1}{2}i}$$

Problem 17.2.15

Find the fixed points.

$$w = \frac{iz + 4}{2z - 5i}$$

Solution

$$\begin{aligned} f(z) &= z \\ w &= \frac{iz + 4}{2z - 5i} = z \\ iz + 4 &= 2z^2 - 5iz \\ 0 &= 2z^2 - 6iz - 4 \\ 0 &= z^2 - 3iz - 2 \quad \implies \quad z = \frac{3i \pm i}{2} = i, 2i \end{aligned}$$

Problem 17.2.19

Find all Linear Fractional Transformations (LFTs) with fixed point(s).

$$z = \pm i$$

Solution

$$w = \frac{az + b}{cz + d}$$

$$w = z \implies cz^2 - (a - d)z - b = 0$$

$$z = i: \quad ci^2 - (a - d)i - b = -(c + b) - (a - d)i = 0$$

$$z = -i: \quad c(-i)^2 - (a - d)(-i) - b = -(c + b) + (a - d)i = 0$$

From the real parts: $c = -b$

From the imaginary parts: $a = d$

$$w = \frac{az + b}{a - bz}, \quad \forall a, b \in \mathbb{C}$$

Problem 17.3.9

Find the LFTs that maps the given three points onto the three given points in the respective order.

$$1, i, -1 \text{ onto } i, -1, -i$$

Solution

$$\begin{aligned} \frac{w - w_1}{w - w_3} \cdot \frac{w_2 - w_3}{w_2 - w_1} &= \frac{z - z_1}{z - z_3} \cdot \frac{z_2 - z_3}{z_2 - z_1} \\ \frac{w - i}{w - (-i)} \cdot \frac{(-1) - (-i)}{(-1) - i} &= \frac{z - 1}{z - (-1)} \cdot \frac{i - (-1)}{i - 1} \\ \frac{w - i}{w + i} \cdot \frac{1 - i}{1 + i} &= \frac{z - 1}{z + 1} \cdot \frac{i + 1}{i - 1} \end{aligned}$$

$$\frac{1 - i}{1 + i} = \frac{(1 - i)^2}{1^2 + i^2} = \frac{-2i}{2} = -i$$

$$\frac{i + 1}{i - 1} = \frac{(i + 1)^2}{i^2 - 1^2} = \frac{2i}{-2} = -i$$

$$\frac{w - i}{w + i} = \frac{z - 1}{z + 1}$$

$$(w - i)(z + 1) = (z - 1)(w + i)$$

$$wz + w - iz - i = zw + iz - w - i$$

$$2w = 2iz$$

$$w = iz$$

$$\boxed{w = iz}$$

Problem 17.3.11

Find the LFT that maps the given three points onto the three given points in the respective order.

$$-1, 0, 1 \text{ onto } -i, -1, i$$

Solution

$$\begin{aligned} \frac{w - w_1}{w - w_3} \cdot \frac{w_2 - w_3}{w_2 - w_1} &= \frac{z - z_1}{z - z_3} \cdot \frac{z_2 - z_3}{z_2 - z_1} \\ \frac{w - (-i)}{w - i} \cdot \frac{(-1) - i}{(-1) - (-i)} &= \frac{z - (-1)}{z - 1} \cdot \frac{0 - 1}{0 - (-1)} \\ \frac{w + i}{w - i} \cdot \frac{-1 - i}{-1 + i} &= \frac{z + 1}{z - 1} \cdot \frac{1}{-1} \end{aligned}$$

$$\frac{-1 - i}{-1 + i} = \frac{(-1 - i)^2}{(-1)^2 + i^2} = \frac{2i}{2} = i$$

$$\frac{w + i}{w - i} \cdot i = \frac{z + 1}{1 - z}$$

$$(w + i)(1 - z) \cdot i = (z + 1)(w - i)$$

$$(w - wz + i - iz) \cdot i = zw - iz + w - i$$

$$iw - i wz - 1 + z = zw - iz + w - i$$

$$iw - i wz - w - zw = -iz - i + 1 - z$$

$$w(i - iz - 1 - z) = -iz - i + 1 - z$$

$$w = \frac{-iz - i + 1 - z}{i - iz - 1 - z}$$

$$\boxed{w = \frac{-(1+i)z + (1-i)}{-(1+i)z + (i-1)}}$$

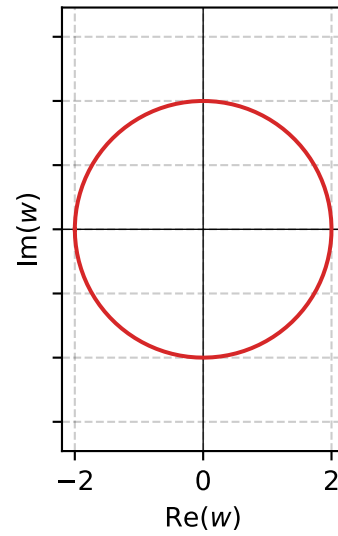
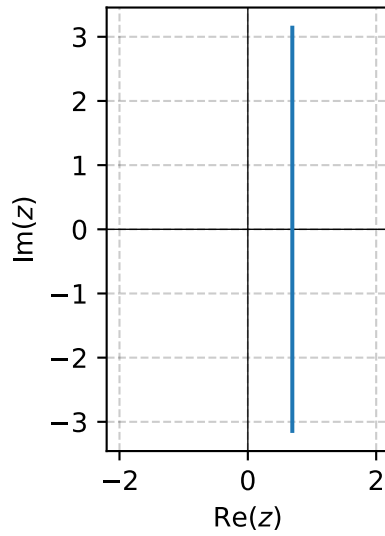
Problem 17.4.1

Find the image of $x = c = \text{const}$, $-\pi < y \leq \pi$, under $w = e^z$.

Solution

$$z = x + iy = c + iy$$

$$w = e^c e^{iy}$$



Problem 17.4.3

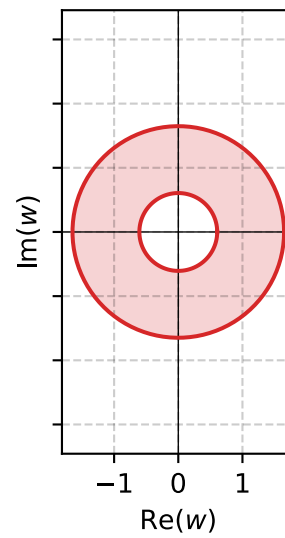
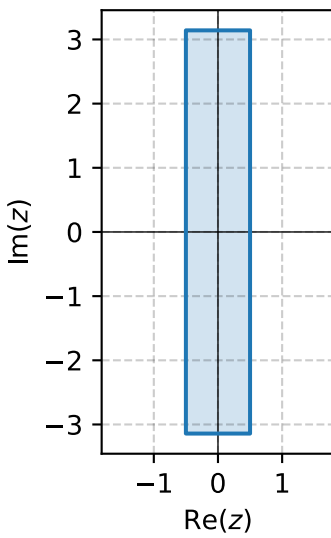
Find and sketch the image of the given region under $w = e^z$.

$$-\frac{1}{2} \leq x \leq \frac{1}{2}, \quad -\pi \leq y \leq \pi$$

Solution

$$z = x + iy$$

$$w = e^x e^{iy}$$



Problem 17.4.5

Find and sketch the image of the given region under $w = e^z$.

$$-\infty \leq x \leq \infty, \quad 0 \leq y \leq 2\pi$$

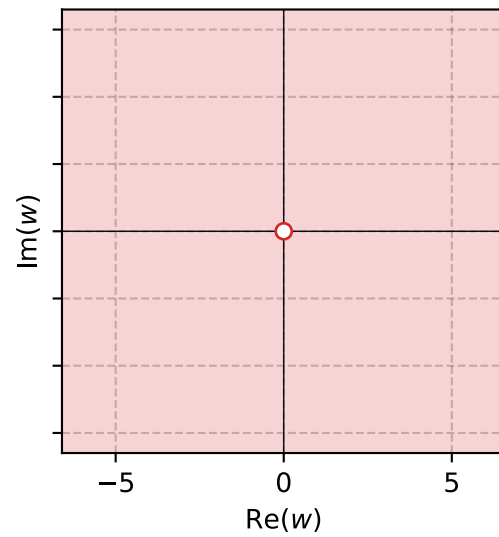
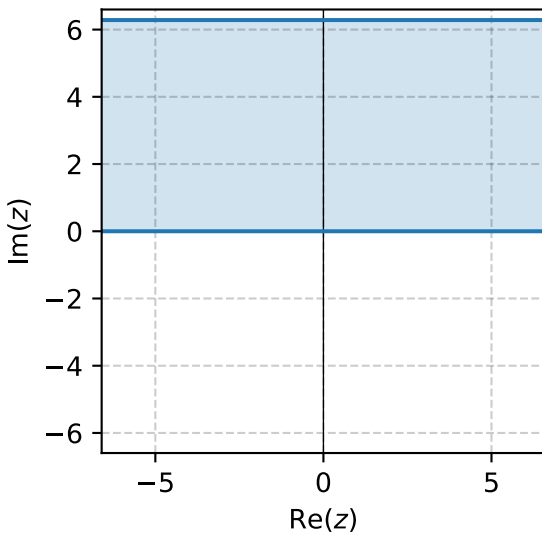
Solution

The initial region is an infinite strip between $0 \leq y \leq 2\pi$,

$$z = x + iy$$

The image fills the entire plane except at the center,

$$w = e^x e^{iy}$$



Problem 17.4.7

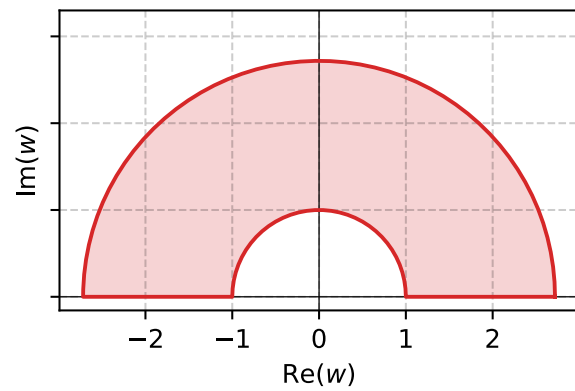
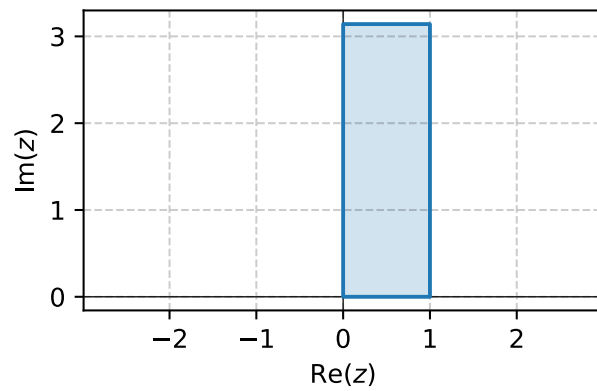
Find and sketch the image of the given region under $w = e^z$.

$$0 < x < 1, \quad 0 < y < \pi$$

Solution

$$z = x + iy$$

$$w = e^x e^{iy}$$



Problem 17.4.11

Find and sketch or graph the image of the given region under $w = \sin z$.

$$0 < x < \frac{\pi}{2}, \quad 0 < y < 2$$

Solution

The region is bounded by 4 straight lines that form a rectangle. Mapping these per 17.4.(2) gives

$$z = x + iy \rightarrow w = \sin x \cosh y + i \cos x \sinh y$$

The left edge ($x = 0, y \in (0, 2)$) maps to a segment on the imaginary axis

$$z = iy \rightarrow w = i \sinh y$$

The top edge ($x \in (0, \frac{\pi}{2}), y = 2$) maps to the upper right quadrant of an ellipse

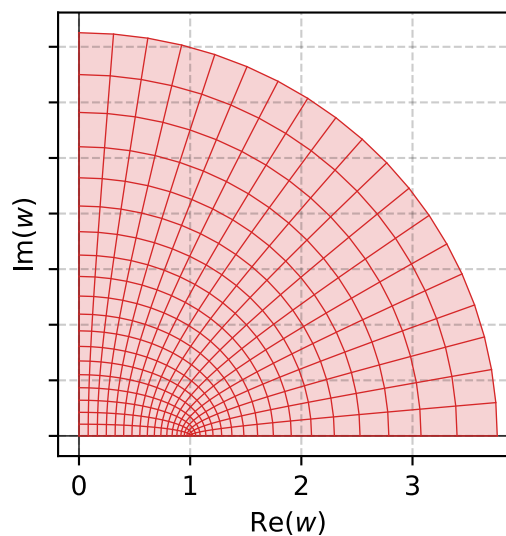
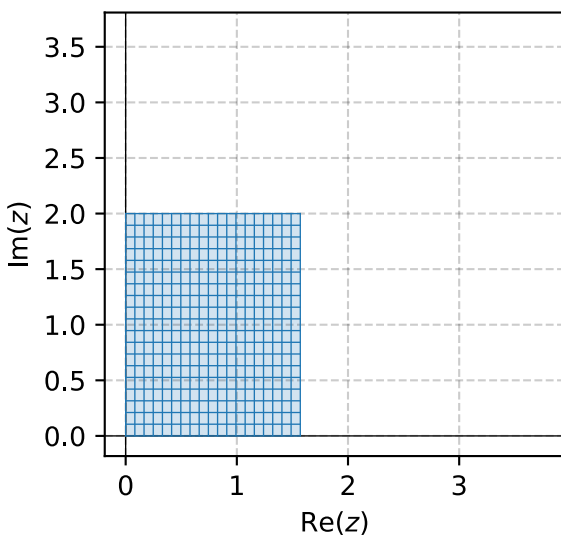
$$z = x + 2i \rightarrow w = \sin x \cosh 2 + i \cos x \sinh 2$$

The right edge ($x = \frac{\pi}{2}, y \in (0, 2)$) maps to a segment on the real axis

$$z = \frac{\pi}{2} + iy \rightarrow w = \cosh y$$

The bottom edge ($x \in (0, \frac{\pi}{2}), y = 0$) maps to a segment on the real axis

$$z = x \rightarrow w = \sin x$$



Problem 17.4.13

Find and sketch or graph the image of the given region under $w = \sin z$.

$$0 < x < 2\pi, \quad 1 < y < 3$$

Solution

The region is bounded by 4 straight lines that form a rectangle. Mapping these per 17.4.(2) gives

$$z = x + iy \rightarrow w = \sin x \cosh y + i \cos x \sinh y$$

The left edge ($x = 0, y \in (1, 3)$) maps to a segment on the imaginary axis

$$z = iy \rightarrow w = i \sinh y$$

Similarly, the right edge ($x = 2\pi, y \in (1, 3)$) maps to the same segment on the real axis

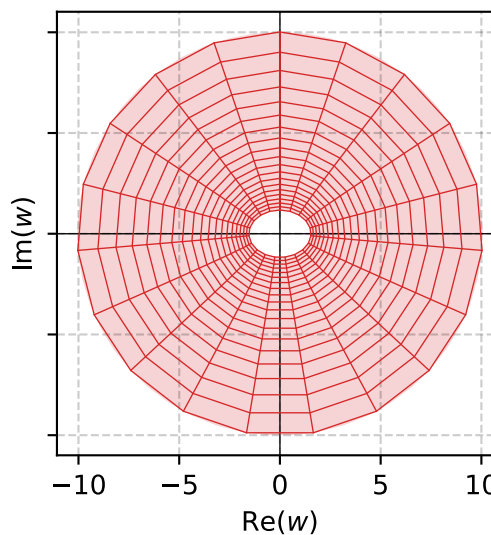
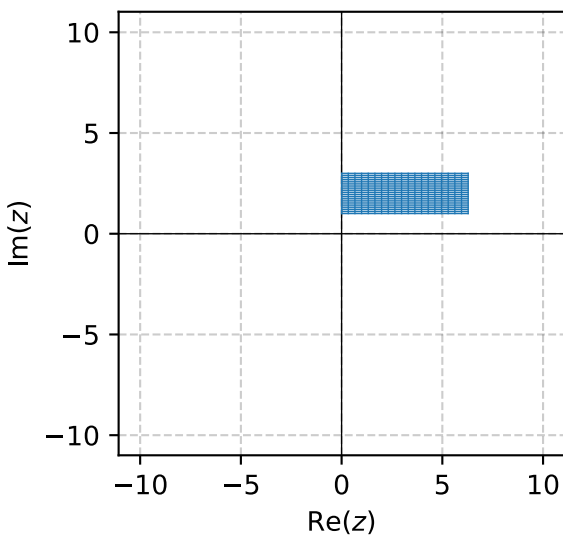
$$z = 2\pi + iy \rightarrow w = \overset{0}{\sin 2\pi} \cosh y + i \overset{1}{\cos 2\pi} \sinh y = i \sinh y$$

The top edge ($x \in (0, 2\pi), y = 3$) maps to a big ellipse

$$z = x + 3i \rightarrow w = \sin x \cosh 3 + i \cos x \sinh 3$$

The bottom edge ($x \in (0, 2\pi), y = 1$) maps to a small ellipse

$$z = x + 1i \rightarrow w = \sin x \cosh 1 + i \cos x \sinh 1$$



Problem 17.4.19

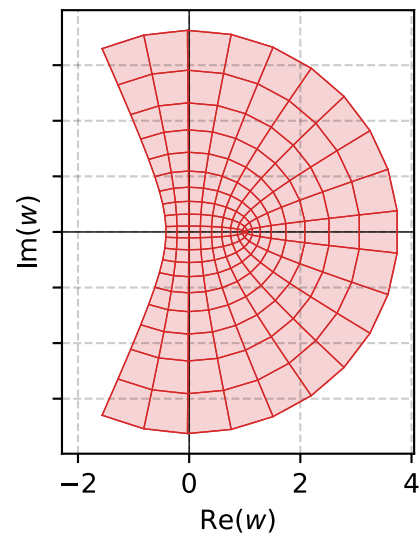
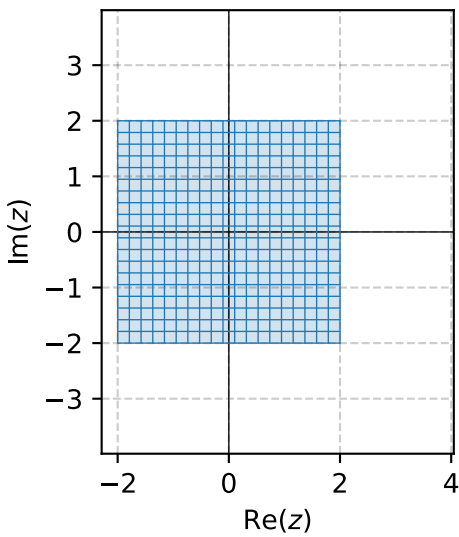
Find the images of the lines mapping $x = c = \text{const}$ under the mapping $w = \cos z$.

Solution

Mapping per 17.4.(3) gives

$$\begin{aligned}
 z = \left(c + \frac{\pi}{2}\right) + iy &\rightarrow w = u + iv \\
 &= \sin\left(c + \frac{\pi}{2}\right) \cosh y + i \cos\left(c + \frac{\pi}{2}\right) \sinh y \\
 &= \cos c \cosh y - i \sin c \sinh y
 \end{aligned}$$

$$\Rightarrow \frac{u^2}{\cos^2 c} - \frac{v^2}{\sin^2 c} = 1 \quad (\text{hyperbola centered at } w=0)$$



Problem 17.4.21

Find and sketch or graph the image of the given region under the mapping $w = \cos z$ directly from **Problem 17.4.11**.

$$0 < x < \frac{\pi}{2}, \quad 0 < y < 2$$

Solution

The region is bounded by 4 straight lines that form a rectangle. Mapping these per 17.4.(2) gives

$$z = x + iy + \frac{\pi}{2} \rightarrow w = \cos x \cosh y - i \sin x \sinh y$$

The left edge ($x = 0, y \in (0, 2)$) maps to a segment on the real axis

$$z = iy + \frac{\pi}{2} \rightarrow w = \cosh y$$

The top edge ($x \in (0, \frac{\pi}{2}), y = 2$) maps to the lower right quadrant of an ellipse

$$z = x + 2i + \frac{\pi}{2} \rightarrow w = \cos x \cosh 2 - i \sin x \sinh 2$$

The right edge ($x = 2, y \in (0, 2)$) maps to a segment on the imaginary axis

$$z = \frac{\pi}{2} + iy + \frac{\pi}{2} \rightarrow w = -\sinh y$$

The bottom edge ($x \in (0, \frac{\pi}{2}), y = 0$) maps to a segment on the real axis

$$z = x + \frac{\pi}{2} \rightarrow w = \cos x$$

